

Quantitative diagonalization of monotone comparison tensors

Asymptotic research note; independently reviewed

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Let $p_1, \dots, p_m$ be nondecreasing $[0, 1]$ -valued functions on a probability space with an ordered finite support. Suppose

$$\mathbb{E} \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Write $\rho = 1/12$ and $d_j = \text{Var}(p_j) - \rho$. These statements hold for arbitrary positive row weights, except where an equal-weight quadrature interpretation is explicitly mentioned.

Theorem 1 (Variance defects). *At most one d_j is negative. If the positive defects in decreasing order are $d_{(1)}, d_{(2)}, \dots$, then*

$$d_{(r)} \leq \frac{r}{4L(r - L + 1)} \leq \frac{1}{r}, \quad L = \left\lfloor \frac{r + 1}{2} \right\rfloor.$$

Consequently $\sum_j d_j \leq 1 + \log m$ and, for $\bar{p} = m^{-1} \sum_j p_j$,

$$\frac{1}{m} \sum_j \mathbb{E}(p_j - \bar{p})^2 = \frac{m - 1}{m^2} \sum_j d_j \leq \frac{1 + \log m}{m}.$$

Proof. The centered Gram matrix is $G = D + \rho \mathbf{1}\mathbf{1}^T$. Two negative defects would give a negative quadratic form on the vector supported on their two positions with entries $1, -1$.

Represent the ordered probability space on $[0, 1]$ by assigning each atom a subinterval of length equal to its weight and extending each p_j constantly. Project these functions onto functions constant on L equal-length subintervals. For one column, let a_ℓ be its oscillation on subinterval ℓ , ignoring

endpoints of measure zero. Monotonicity gives $\sum_{\ell} a_{\ell} \leq 1$. Its squared projection error is at most

$$\frac{1}{4L} \sum_{\ell} a_{\ell}^2 \leq \frac{1}{4L},$$

by the elementary variance bound for a variable in an interval.

Choose r columns with defects at least $\delta = d_{(r)} > 0$. Their Gram matrix is at least δI_r . The projected centered columns span a space of dimension at most $L - 1$. By orthogonal projection, or the variational principle for singular values, their total squared projection error is at least $(r - L + 1)\delta$. Its upper bound is $r/(4L)$, proving the estimate. The harmonic-sum bound follows. Finally direct expansion of the centered Gram matrix gives the displayed identity for the average distance to $\bar{p}$. $\square$

Lemma 1 (Elementary symmetric means). *For $x_1, \dots, x_r \in [0, 1]$, put $t = r^{-1} \sum_i x_i$ and $V = r^{-1} \sum_i (x_i - t)^2$. For $2 \leq s \leq r$,*

$$0 \leq t^s - \frac{e_s(x_1, \dots, x_r)}{\binom{r}{s}} \leq \frac{\binom{s}{2}}{r-1} V.$$

Proof. Replace any pair x, y by two copies of $(x + y)/2$. This increases e_s by $((x - y)^2/4)e_{s-2}$ of the remaining $r - 2$ coordinates, and decreases V by $(x - y)^2/(2r)$. Since all remaining coordinates are in $[0, 1]$,

$$0 \leq \Delta \frac{e_s}{\binom{r}{s}} \leq \frac{\binom{s}{2}}{r-1} (-\Delta V).$$

Repeated pair averaging can be chosen to converge to the constant vector $(t, \dots, t)$: for example, always average a smallest and a largest entry; unless $V = 0$, its decrease is at least $V/(2r)$. Telescoping and continuity prove both inequalities. $\square$

Theorem 2 (A scalar approximate rule from many columns). *Let I be any set of $r \geq 2$ columns and put $t = r^{-1} \sum_{j \in I} p_j$ and $\Delta = \sum_{j \in I} d_j$. Then $\Delta \geq 0$, $\mathbb{E}t = 1/2$, and simultaneously for $2 \leq s \leq r$,*

$$0 \leq \mathbb{E}t^s - \frac{1}{s+1} \leq \binom{s}{2} \frac{\Delta}{r^2}.$$

In particular, remove R columns with largest defects, where $1 \leq R \leq m - 2$, and average the remaining $r = m - R$ columns. Then

$$0 \leq \mathbb{E}t^s - \frac{1}{s+1} \leq \frac{\binom{s}{2}}{Rr} \quad (2 \leq s \leq r).$$

For $R = \lfloor m/2 \rfloor$, this is $O(s^2/m^2)$ through a degree of order m . If the original rows have equal weights, the values of t form an equal-weight scalar approximate quadrature rule on the same number of nodes.

Proof. At each row apply Lemma 1 to the selected columns. The expectation of their normalized elementary symmetric polynomial is exactly $1/(s+1)$. Their mean squared dispersion is

$$\mathbb{E}V = \frac{1}{r} \sum_{j \in I} \mathbb{E}p_j^2 - \mathbb{E}t^2 = \frac{r-1}{r^2} \Delta,$$

so $\Delta \geq 0$ and the first estimate follows. Every remaining positive defect after deletion is at most $1/R$ by the preceding theorem; negative defects only decrease Δ . Hence $\Delta \leq r/R$, yielding the second estimate. $\square$

Lemma 2 (A projection adapted to a small mean). *Let q be a nonincreasing $[0, 1]$ -valued function on $[0, 1]$ with $\int q = \alpha > 0$. There is a partition into L intervals, depending only on α and L , such that its orthogonal step projection Pq satisfies*

$$\|q - Pq\|_2^2 \leq \frac{4\alpha}{L(1+\alpha)}.$$

The same partition works for every such q .

Proof. Partition $u = t/(\alpha + t)$ into L equal increments from 0 to $1/(1+\alpha)$, and pull the partition back to $[0, 1]$. For a step $h_s(t) = \mathbf{1}_{[0,s]}(t)$ with s in a partition interval $[a, b]$, its squared projection error is

$$\|h_s - Ph_s\|_2^2 = \frac{(s-a)(b-s)}{b-a} \leq s-a \leq \frac{(\alpha+s)^2}{\alpha L(1+\alpha)}.$$

The last inequality follows from $s-a = (\alpha+s)(\alpha+a)(u(s)-u(a))/\alpha$. Write $q = q(1) + \int h_s d\nu(s)$ with ν a positive measure. Then $\nu([0, 1]) \leq 1$ and $\int s d\nu(s) \leq \alpha$. The triangle inequality in L^2 gives

$$\|q - Pq\|_2 \leq \frac{\int (\alpha+s) d\nu(s)}{\sqrt{\alpha L(1+\alpha)}} \leq 2\sqrt{\frac{\alpha}{L(1+\alpha)}}.$$

This proof also covers finite step functions and atoms exactly. $\square$

Corollary 1 (Endpoint-tilted diagonalization). *Fix a set S of k columns and reweight the rows by*

$$d\mathbb{P}_S = (k+1) \prod_{i \in S} p_i d\mathbb{P}.$$

The remaining columns have mixed moments $\mu_s^{(k)} = (k+1)/(k+s+1)$. Define their variance defects by

$$d_j^{(k)} = \mathbb{E}_S p_j^2 - \frac{k+1}{k+3}.$$

The r th largest positive defect satisfies

$$d_{(r)}^{(k)} \leq \frac{16}{(k+2)r}.$$

After deleting $R \geq 1$ remaining columns with largest defects and averaging the other $r = m - k - R \geq 2$ columns to obtain t , one has, simultaneously for $2 \leq s \leq r$,

$$0 \leq \mathbb{E}_S t^s - \frac{k+1}{k+s+1} \leq \frac{16 \binom{s}{2}}{(k+2)Rr}.$$

In particular, when k, R, r are all of order m , the absolute monomial errors are $O(s^2/m^3)$ through degree of order m .

Proof. Rows assigned zero new weight can simply be omitted. Mixed moments follow by adjoining S to any selected set of remaining columns. The common mean is $(k+1)/(k+2)$ and the off-diagonal covariance is

$$\rho_k = \frac{k+1}{(k+2)^2(k+3)}.$$

Apply Lemma 2 to $q_j = 1 - p_j$, of common mean $\alpha = 1/(k+2)$, using the ordered probability coordinate for the new row weights. A subfamily of u columns whose defects are at least $\delta > 0$ has Gram matrix at least δI_u . The projection argument therefore gives

$$(u - L + 1)\delta \leq \frac{4u\alpha}{L(1+\alpha)}.$$

Choose $L = \lfloor (u+1)/2 \rfloor$ and use $L(u - L + 1) \geq u^2/4$ to obtain $\delta \leq 16\alpha/u$. Finally Lemma 1 and the exact dispersion identity from Theorem 2 apply unchanged to the beta moments. The sum of the retained defects is at most $16r/((k+2)R)$, proving the claim. $\square$

For example, any two columns left after the deletion in Theorem 2 satisfy $\mathbb{E}(p_i - p_j)^2 = d_i + d_j \leq 2/R$. None of these conclusions assumes strict increase or rules out flat columns. The moment error in Theorem 2 is an absolute error for monomials. It does not supply a small error for arbitrary degree- r polynomials, whose monomial coefficients can be large. In particular this note does not claim the universal quadratic lower bound on the number of rows, or an exact scalar quadrature replacement.